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# Broadband power efficient Class E amplifiers with a non-linear CAD model of the active MOS device

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**SUMMARY**

This paper describes how broadband power efficient Class E amplifiers can be designed which are capable of efficiencies approaching 100% over 35% fractional bandwidths. The paper contains a brief review of switching Class E amplifiers, which are normally narrow band, and describes how these can be made broadband. As an example, a broadband 130 to 180 MHz r.f. amplifier has been designed, built and tested and the results are reported here. To enable an accurate computer simulation of the r.f. power amplifier to be made, a non-linear CAD model of the active device (an r.f. power MOSFET) is developed. CAD techniques for matching the non-linear input impedance of the MOSFET are also presented. Experiment and theory are compared and show close correlation.

## 1 Introduction

Modern portable radios are required to operate over a large number of channels for long periods from a small battery pack. This paper describes techniques whereby broadband power-efficient Class E amplifiers, with a passband ripple of less than 1 dB, can be designed and built. The new amplifiers are capable of operating over 35% fractional bandwidths with efficiencies approaching 100%. As an example, a 130 to 180 MHz Class E amplifier has been designed and built using these techniques. At v.h.f. frequencies the efficiency reduces due to the non-ideal switching properties of r.f. power devices. A large signal model for an r.f. MOS device is therefore developed, based on d.c. and small signal *S*-parameter measurements, to allow more detailed analysis of the Class E amplifier. The non-linearities incorporated in the model include the non-linear transconductance of the device, including the reverse biased diode inherent in the MOS structure, and non-linear feedback and output capacitors. The technique used to develop this model can be applied to other non-linear devices. Close correlation is shown between experimental and CAD techniques at 150 MHz. CAD techniques for rapidly matching the input impedance of the non-linear model are also presented.

## 2 Switching Amplifiers

High efficiency amplification is usually achieved by using a switching amplifier where the switch dissipates no power and all the power is dissipated in the load. An ideal switching device dissipates no power because it has no voltage across it when it is on, no current flowing through it when it is off, and zero transition times.

In real switching amplifiers there are three main loss mechanisms:

(i) The non-zero on-resistance of the switching device.
(ii) The simultaneous presence of large voltages and currents during the switching transitions.
(iii) The loss of the energy stored in the parasitic shunt capacitance of the switching device at switch-on.

The losses caused by the on-resistance can be reduced by ensuring that the on-resistance is considerably less than the load resistance presented to the switching device.

The switching transition losses can be reduced by choosing a device with a fast switching time. Efficiency can be further increased if the overlap of the voltage and current waveforms can be reduced to minimize the power losses during the switching transitions.

The loss of the energy stored in the shunt capacitance at switch-on ($\frac{1}{2}CV^2$) can be reduced by choosing a device with a low parasitic shunt capacitance. At v.h.f. even a small parasitic shunt capacitance can result in large losses of energy. The requirement to discharge this capacitor at switch-on also imposes secondary stress on the switching device.

## 3 Class E Amplifiers

The Class E amplifier proposed by the Sokals[1-3] and further analysed by Raab[4,5] is designed to avoid discharging the shunt capacitance of the switching device and to reduce power loss during the switching transitions. This is achieved by designing a load network for the amplifier, which determines the voltage across the switching device when it is off, to ensure minimum losses.

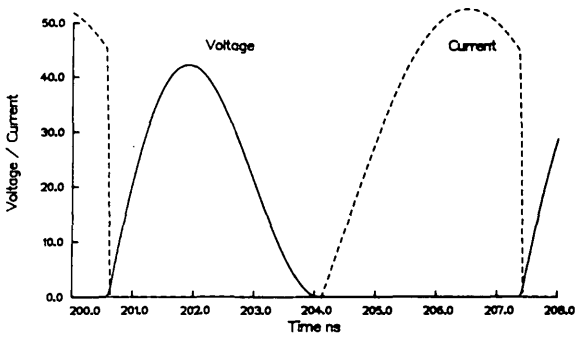



**Fig. 1.** Class E amplifier voltage and current waveforms.


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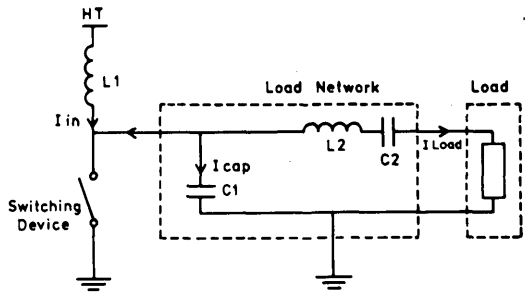



**Fig. 2.** Basic Class E amplifier.

Typical Class E amplifiers waveforms are shown in Fig. 1 where the design criteria for the voltage are that it:

(i) rises slowly at switch-off;
(ii) falls to zero by the end of the half-cycle;
(iii) has a zero rate of change at the end of the half-cycle.

A slow rise in voltage at switch-off reduces the power lost during the switch-off transition. Zero voltage across the switching device at the end of the half-cycle ensures that there is no charge stored in the parasitic shunt capacitance when it turns on, so that no current is discharged through the device. Zero rate of change at the end of the half-cycle reduces power loss during a relatively slow switch-on transition by ensuring that the voltage across the switching device remains at zero while the device is switching on.

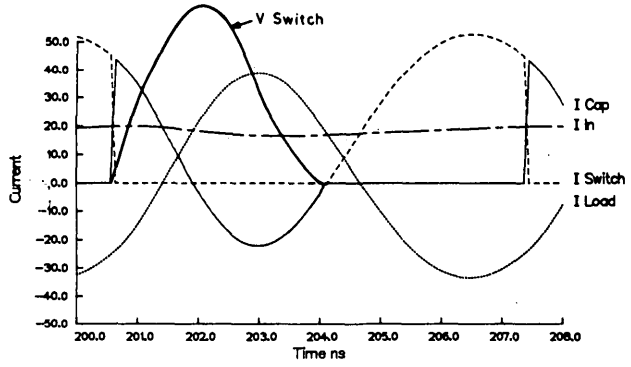



**Fig. 3.** Class E current waveforms ($F_c$ = 147 MHz).

The circuit developed by the Sokals (Fig. 2) uses a single switching device (BJT or FET) and a load network consisting of a series tuned $LC$ network ($L_2$,$C_2$), an r.f. choke ($L_1$), and a shunt capacitor ($C_1$), which may be partly or wholly made up of the parasitic shunt capacitance of the switching device.

The r.f. choke ($L_1$) is sufficiently large to provide a constant input current from the power supply. The series $LC$ circuit ($L_2$,$C_2$) is tuned to a frequency lower than the operating frequency and can be considered, at the operating frequency, as a series tuned circuit in series with an extra inductive reactance. The tuned circuit ensures a substantially sinusoidal load current (Fig. 3) and the inductive reactance causes a phase shift between this current and the fundamental component of the applied voltage. The difference between the constant input current and the sinusoidal load current flows through the switching device when it is on and through the shunt capacitor ($C_1$) when it is off. The capacitor/switch current is therefore also sinusoidal; however, it is now 180 degrees out of phase with respect to the load current and contains a d.c. offset to allow for the current flowing through the r.f. choke ($L_1$).

As the voltage across the switching device when it is off is the integral of the current through the shunt capacitor ($C_1$), the phase shift introduced by the $LC$ circuit adjusts the point at which the current is diverted from the switch to the capacitor. This ensures that the voltage waveform (Fig. 1) meets the criteria for Class E operation by integrating the correct portion of the offset sinusoidal capacitor current. This point is determined by ensuring that the integral of the capacitor current over the half-cycle is zero and that the capacitor current has dropped to zero by the end of the half-cycle.

Due to the fact that the $LC$ series tuned circuit is tuned to a frequency which is lower than the operating frequency, the conventional Class E amplifier has a highly frequency dependent amplitude characteristic (Fig. 4). It is this change of impedance which prevents optimum Class E operation from being achieved over a wide bandwidth.

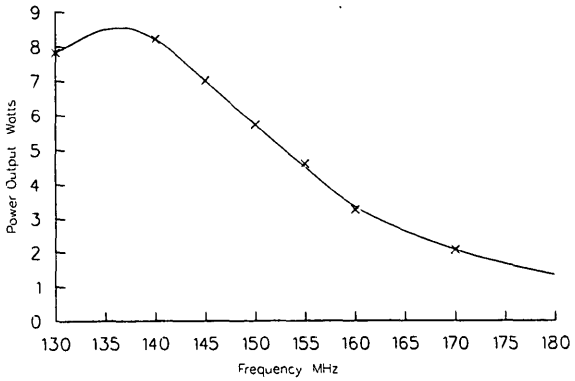



**Fig. 4.** Narrow band Class E amplifier frequency response.

It seems that no work has been published on practical broadband Class E amplifier designs, although Raab briefly mentions the concept on page 243 of Ref. 5 where he shows that high efficiency can be maintained over a large bandwidth. However, his circuit still produces a frequency dependent amplitude characteristic. A novel technique and circuit were therefore developed by the authors to allow a Class E amplifier to operate efficiently over a 35% fractional bandwidth with a flat amplitude response.

## 4 Broadband Class E Amplifiers

To enable the design of broadband Class E amplifiers a closer examination of the voltages and currents of the narrow band amplifier is required (Fig. 3).

As the voltage across the switch is defined by the integral of the current flowing through the shunt capacitor ($C_1$), and as the a.c. component of this current also flows through the series $LC$ circuit ($L_2$,$C_2$) when the switch is off, then the load angle of the series tuned circuit defines the optimum angle for producing the correct voltage waveform. This load angle defines the phase shift between the fundamental components of the voltage across the switch and the current flowing through the series tuned circuit ($L_2$,$C_2$). In the basic Class E amplifier circuit the harmonic impedance of the series tuned circuit is assumed to be high due to its Q. The value of the shunt capacitor ($C_1$) must also be correct to produce the correct voltage when the switch is off and to satisfy the steady state conditions. The load angle of the total network is also therefore important.



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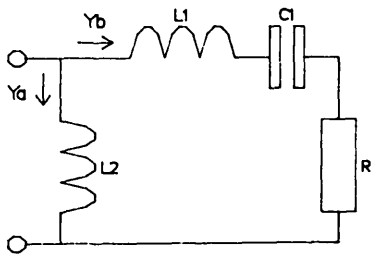



**Fig. 5.** Broadband load angle network.

Simulation of an ideal narrow band circuit (Fig. 2) shows that the load angle of the off-tuned series tuned circuit should be 50 degrees and the angle of the total circuit should be 33 degrees. This is confirmed by Raab in Ref. 5.

If the load network is designed without incorporating a shunt capacitor a simple broadband network can be designed. This should be designed with a greater load angle (50 degrees), which reduces to the required 33 degrees when a shunt capacitor is added. The slope in susceptance with frequency caused by this capacitor is removed as described later.

A circuit capable of presenting a constant load angle over a very large bandwidth is shown in Fig. 5 and its susceptance diagram in Fig. 6. The circuit consists of a low Q series tuned circuit in parallel with an inductor. At the resonant frequency of the tuned circuit the slope of its susceptance curve is designed to cancel the slope of the susceptance curve of the inductor. This allows the circuit to maintain a constant susceptance over a wide bandwidth. An analysis of this circuit is given in the Appendix and it is shown that optimum flatness can be achieved when:

$$L_2 = R/\omega\tan 50°$$
$$C_1 = 2L_2/R^2$$
$$L_1 = 1/\omega^2 C_1$$

The load network impedance at the harmonics should be purely reactive to ensure no losses in the load network.

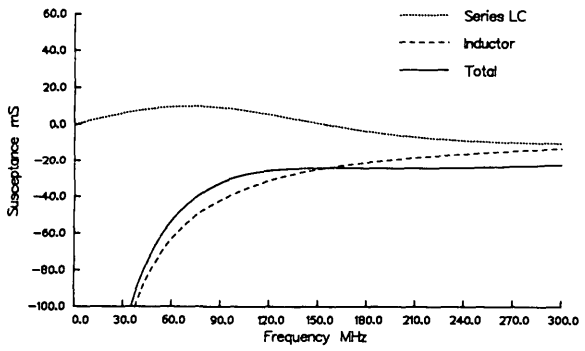



**Fig. 6.** Susceptance of broadband circuit.

The impedance should also be fairly high to ensure that the integral of the current through the shunt capacitor, and hence the current through the capacitor, should be similar to that in the narrow band circuit to meet the criteria for Class E operation. It should also be high to avoid harmonic power being dissipated in the on-resistance of the switching device.

To reduce the power output at the harmonics this circuit was combined with a broadband matching network and a third order bandpass filter to produce a circuit which presented a load angle of 50 degrees over the band 130 MHz to 180 MHz. The filter was based on a Chebyshev low-pass filter design, obtained from Zverev,[6] which had been converted to a bandpass filter. The matching network was arranged to increase the output power of the amplifier by decreasing the load presented to the device. The final network was designed to deliver 12 W into a 50 Ω load using a 12 V power supply. This network can be considered as the broadband equivalent of the off-tuned series *LC* circuit of the simple Class E amplifier, which presents a load angle of 50 degrees at the operating frequency.

When the shunt capacitor is added to this network a slope in susceptance is introduced due to the capacitor's frequency dependence. A slope was therefore placed on the impedance curve of the network using a CAD a.c. optimization package so that when the shunt capacitance was included in the network it presented a constant 33 degree load angle and a constant magnitude of input impedance of 12 Ω over the band 130 MHz to 180 MHz. The impedance of the load network without the capacitor now has a load angle of 50 degrees in the middle of the band which slightly increases at higher frequencies and slightly reduces at lower frequencies.

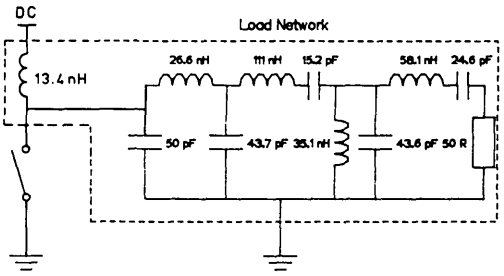



**Fig. 7.** Broadband load network.

During a.c. optimization it was found that a number of components could be removed without degrading the performance. The final network and its impedance are shown in Figs. 7 and 8. The broadband amplifier was then simulated in the time domain (Figs 9 and 10) using a switch with the following characteristics:

(i) 1 Ω on resistance.
(ii) 1 MΩ off resistance.
(iii) 1 ns switching time.

The simulation showed that the amplifier, with a 12 V supply, was capable of delivering 12 W (11 dBW) into a

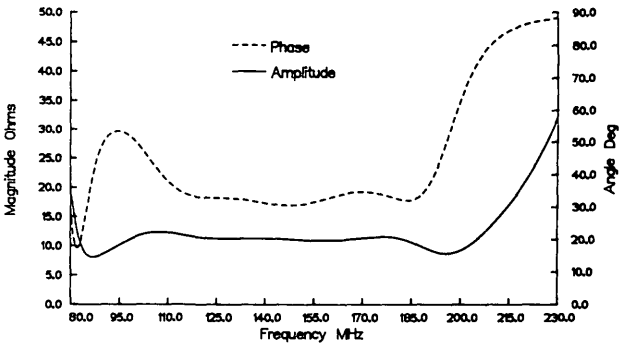



**Fig. 8.** Broadband load network impedance.



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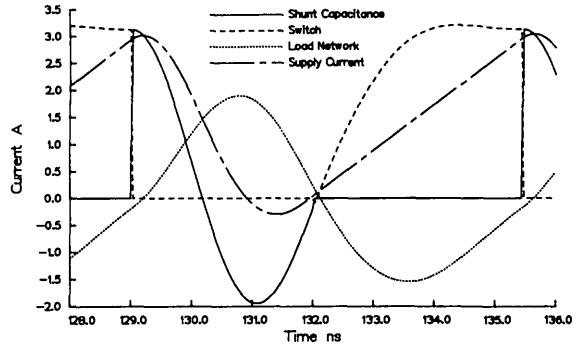



**Fig. 9.** Broadband Class E amplifier current waveforms.

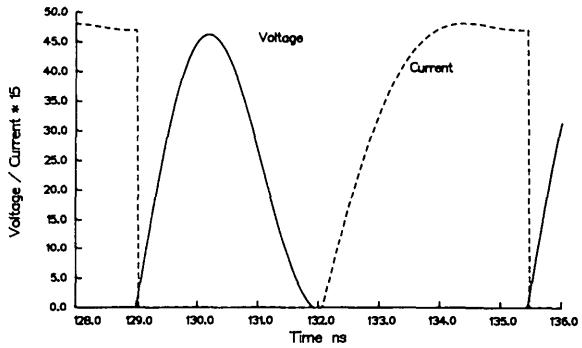



**Fig. 10.** Broadband Class E amplifier voltage and current waveforms.

50 Ω load with 85% efficiency over the band 130 MHz to 180 MHz (Figs 11, 12). A Discrete Fourier Transform showed that 2nd, 3rd and 4th order harmonics were all better than 45 dB below the fundamental.

The current waveforms (Figs 9, 10) are different from those for the simple Class E amplifier (Figs 1, 3), with the exception of the current through the shunt capacitor while the switching device is off. As this is the same, the voltage across the device (Fig. 10) is the same as for the simple Class E amplifier and the criteria for maximum efficiency described earlier are met. The r.f. choke is now part of the load network and therefore the input current is now an asymmetric sawtooth.

## 5 Measurements

A broadband amplifier was constructed and the impedance of the load network was checked with a network analyser. A Mullard MOSFET (1122BLY) was used as the switching

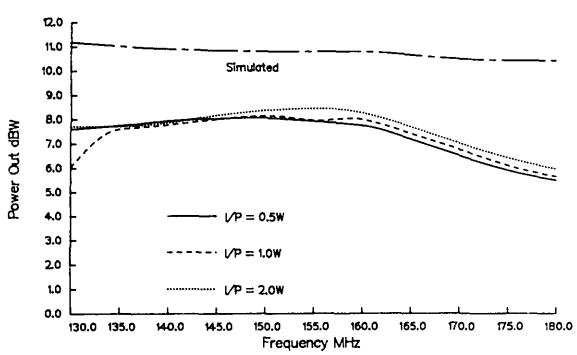



**Fig. 11.** Class E broadband amplifier. Graph of power out vs frequency.

device; this FET has a parasitic shunt capacitance of approximately 35 pF, so an additional 22 pF trimmer was placed in parallel with the device to achieve the required capacitance. The measured results (Figs 11, 12) show that the output power remained fairly constant over the band at approximately half the value for the simulated amplifier. The efficiency remained fairly constant at approximately 60% and a power gain of 10 dB was achieved with a 0·5 W drive power. The input matching network of the constructed amplifier was not broadband and was adjusted for a perfect match using a directional coupler at each frequency measurement. A broadband input matching network could be designed to achieve a complete broadband amplifier.

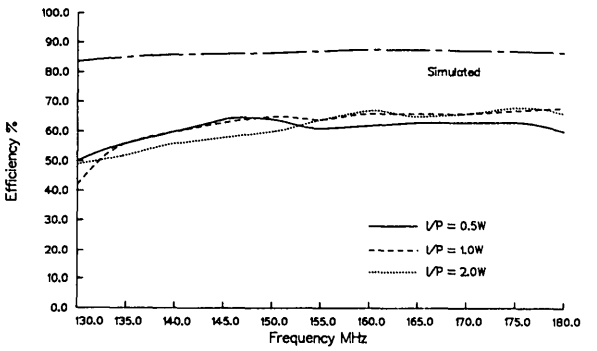



**Fig. 12.** Class E broadband amplifier measurements. Graph of efficiency vs frequency.

## 6 Non-linear Modelling

To enable more accurate analysis of the experimental amplifier a non-linear model of the active MOS device was developed. The circuit used for the large signal model is shown in Fig. 13 where the component values are determined by measurements from an actual device, the Mullard 1122BLY.

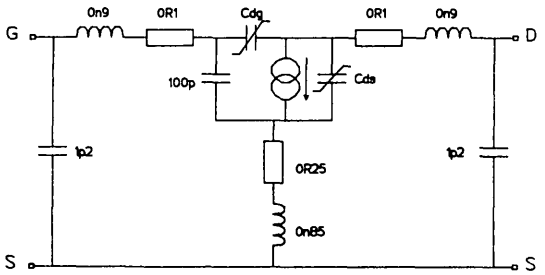



**Fig. 13.** MOSFET model.

A test jig was built for the Mullard 1122BLY and the d.c. characteristics of the device were measured on a curve tracer. The measured output current versus input voltage ($I_d$ vs $V_{gs}$) characteristics were modelled using an equation which incorporates:

(i) a threshold voltage,
(ii) a non-linear region up to 5 V,
(iii) a linear region above 5 V.

The measured output current versus output voltage ($I_d$ vs $V_{ds}$) characteristics were modelled using an equation



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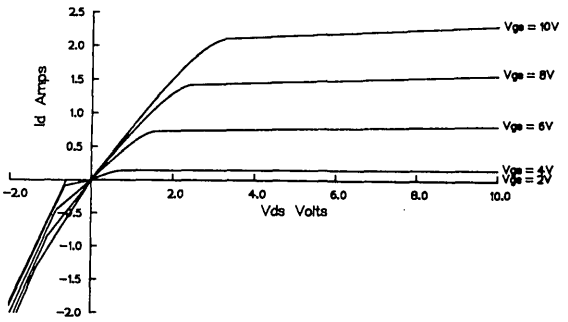



**Fig. 14.** MOS model static characteristics. Graph of $I_d$ vs $V_{ds}$.

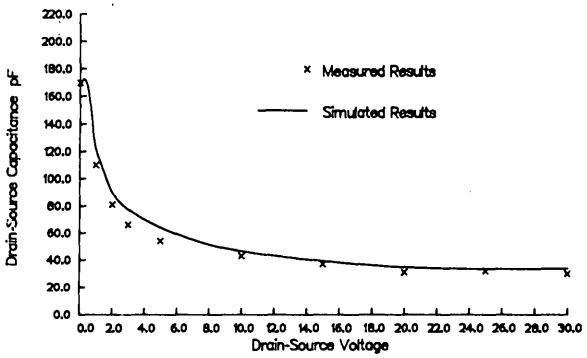



**Fig. 16.** MOS model dynamic characteristics: drain–source capacitance.

which incorporates:

(i) reverse breakdown (due to the parasitic diode formed by the p-type channel and the n-type drain),
(ii) a linear region below pinch-off,
(iii) a region with a small slope (due to the effective output impedance of the current source).

The static characteristics of the model were measured in a simulated test circuit and are shown in Fig. 14.

The capacitor values for the model were obtained from a.c. measurements of the 1122BLY device at 145 MHz. The *S*-parameters of the FET were measured with a network analyser for various gate–source and drain–source voltages. These *S*-parameters were converted to *Y*-parameters and the capacitor values were obtained from the *Y*-parameters using the following equations:

$$C_{dg} = \frac{-\mathrm{Im}(Y_{12})}{\omega}$$

$$C_{ds} = \frac{\mathrm{Im}(Y_{22})}{\omega} - C_{dg}$$

$$C_{gs} = \frac{\mathrm{Im}(Y_{11})}{\omega} - C_{dg}$$

It should be noted that these equations are approximate as they do not take the lead inductors into account. However, the measurements were performed at low frequencies to reduce the effect of the lead inductances. Further, these equations are only correct as long as the device series resistance is small. The gate–source capacitor was found to have an almost constant value of 100 pF independent of $V_{ds}$ or $V_{gs}$; however, the other two capacitors were found to be non-linear. This is confirmed in Ref. 7.

The measured and simulated feedback and output capacitors are shown in Figs 15 and 16 and show close correlation. The drain–gate feedback capacitance was assumed to be independent of the drain–source voltage and the drain–source output capacitance was assumed to be independent of the drain–gate voltage. Measurements indicated that this was a reasonable approximation.

The capacitance values obtained from the *S*-parameter measurements of the 1122BLY give the small signal change in charge with respect to the voltage across the capacitor. As the total charge in the capacitor needs to be defined as a function of the voltage, it is the integral of the measured curve which is specified in the model. The constant of integration is the charge on the capacitor when there is no voltage across it and is set to zero. These functions are entered into the model as tables because no suitable equation has been found which would follow the required curves for both positive and negative voltages.

An initial test of the model was made by putting it into a simulated switching circuit with a resistive load and comparing the results with those obtained from an experimental jig. The source of the FET was connected to ground and the drain was connected to a positive 5 V power supply via a 9·4 Ω resistor. (A 2 nH parasitic lead inductance was incorporated in the simulation.) The drain of the FET was connected to a 300 ps sampling oscilloscope which presented 50 Ω across the drain and source of the FET. The circuit was driven from a 145 MHz sine wave generator with a 50 Ω output impedance. A matching network was used at the gate so that the circuit presented a 50 Ω impedance to the sine wave generator. The matching network predicted by the model was found to be similar to that required by an 1122BLY when used in a constructed switching circuit. The drain waveform obtained from the simulation (Fig. 17) is similar to that obtained from the constructed circuit (Fig. 18). Both of

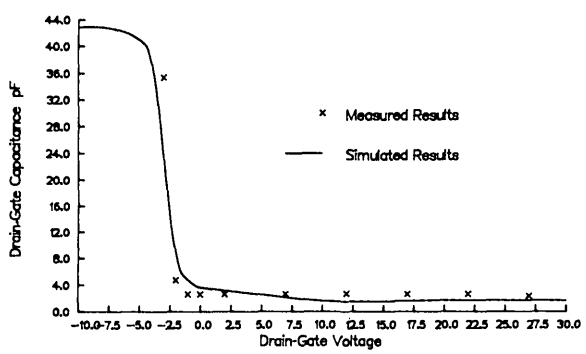



**Fig. 15.** MOS model dynamic characteristics: drain–gate capacitance.

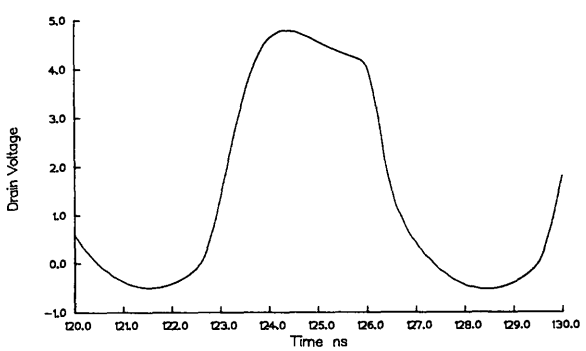



**Fig. 17.** MOSFET model switching waveform ($V_{supply}$ = 5 V, $V_{bias}$ = 4 V, input power = 0·8 W, $R_{load}$ = 9·4//50 Ω).



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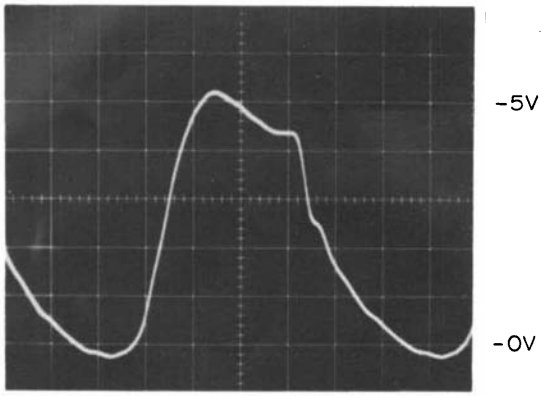



**Fig. 18.** 1122BLY switching waveform ($V_{supply}$ = 5 V, $V_{bias}$ = 4 V, input power = 0·8 W, $R_{load}$ = 9·4//50 Ω), X-axis 1 ns/div, Y-axis 1 V/div.

these observations suggest that the model is accurately predicting the FET characteristics under non-linear operating conditions at 145 MHz.

## 7 CAD Design of Input Matching Networks

Matching the non-linear input impedance of the simulated circuit in the time domain takes a large number of iterations and considerable time (many days); a rapid technique for matching was therefore developed. The drain and the source of the FET were connected to the broadband circuit and the gate was connected to a sinusoidal voltage source set to the correct operating voltage. A CAD Fourier transform was performed to find out the amplitude and phase of the fundamental component of the input voltage and current. The Fourier transform was performed within the CAD package by multiplying the voltage and current by quadrature components and integrating these functions over one period using a current source, switch, and capacitor. This ensured high accuracy in the Fourier transform as the CAD package adjusts the time intervals to maintain accuracy. The fundamental impedance was then calculated and a matching network designed. When the network was incorporated into the simulation an almost perfect match was achieved. Figure 19 shows the gate voltages and currents and the voltages and currents at the input of the matching network. It is interesting to note that the gate voltage and the voltage at the input to the matching network are similar in magnitude even though the impedances are very different. This is because the device input impedance was found to be (1 + 7j) Ω which can be modelled as a 50 Ω resistor in parallel with a capacitor. As the matching network is assumed to be lossless the voltages would be the same to ensure power conservation.

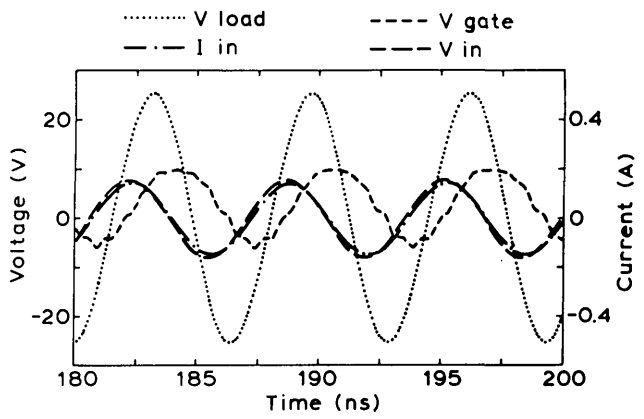



**Fig. 19.** Broadband Class E amplifier operating at 155 MHz ($R_{load}$ = 50 Ω).

## 8 Simulations of the Broadband Amplifiers

Simulations of the complete broadband amplifier were performed with the non-linear model at 155 MHz. The input power to the amplifier was 600 mW and the output power was 6·25 W (Fig. 19). This shows close correlation with the measured results (Figs 11 and 12). The simulated efficiency at 155 MHz using the new non-linear model was 75% and the experimental efficiency was 65%. This shows a lower and more accurate prediction of efficiency than was predicted using the resistive switch model described in Section 4, the results of which are shown in Fig. 12. The measured drain–source voltage can be compared with the simulated voltage and both show a peak voltage swing of 35 V and a similar shape (Figs 20(a),(b)).

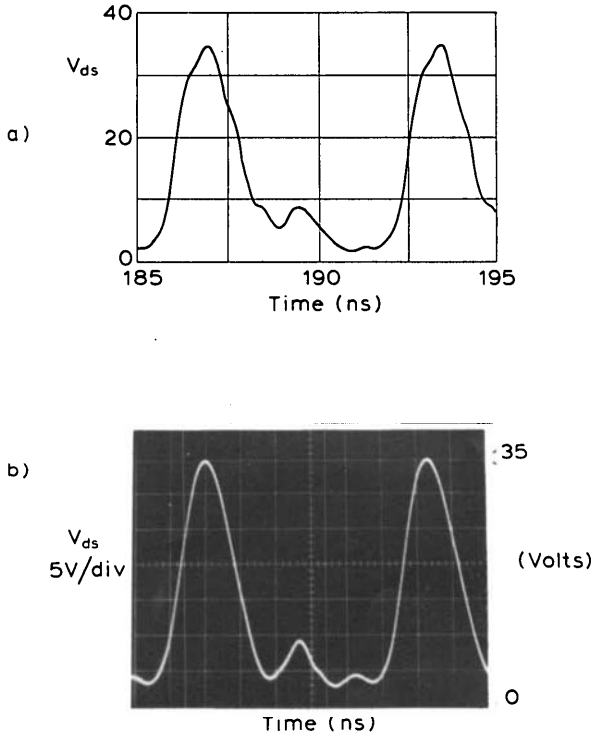



**Fig. 20.** Drain–source voltage at 155 MHz. (a) Simulated. (b) Measured.

## 9 Conclusions

The operation of Class E amplifiers has been studied and a technique has been developed which enables amplifiers to be designed which are capable of maintaining very efficient Class E operation over a 35% bandwidth. Initial broadband amplifiers have been shown to be capable of 60% efficiency over a 50 MHz bandwidth at 150 MHz. However, much higher efficiencies are possible if better switching devices can be found. A large signal model of a MOSFET has been developed to allow more accurate simulation of practical switching amplifiers. The model includes non-linearities in $g_m$ due to the gate-source voltage, the drain–source voltage and the parasitic reverse diode. The model also incorporates non-linear drain–gate and drain–source capacitors, and parasitic package components. Simulations incorporating the model have shown close correlation with experiment. A novel CAD technique for rapidly matching the non-linear input impedance has also been presented.



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## 11 References

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## 12 Appendix: Load Angle Network

A circuit capable of presenting a constant load angle over a broad bandwidth is analysed. The circuit consists of an inductor in parallel with a low $Q$ series tuned circuit.

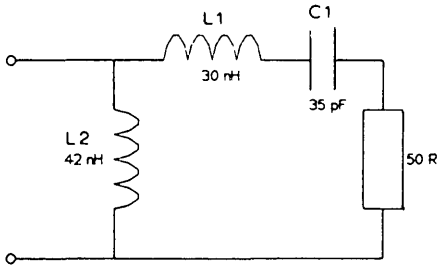



**Fig. 21.** Broadband load angle network.

The load angle is set by the ratio of the impedance of the inductor $L_2$ to the resistor $R$. In order to maintain this load angle over a broad band the slope of the susceptance of $L_2$ is cancelled by the slope of the resonant circuit $L_1,C_1$.

$$Y_a = \frac{-j}{\omega L_2}$$

$$Y_b = \frac{1}{R+j\left(\omega L_1 - \frac{1}{\omega C_1}\right)} = \frac{\omega^2 C_1^2 R - j\omega C_1(\omega^2 C_1 L_1 - 1)}{(\omega C_1 R)^2 + (\omega^2 C_1 L_1 - 1)^2}$$

At the resonant frequency of $C_1$ and $L_1$

$$\omega^2 C_1 L_1 - 1 = 0$$

Therefore

$$Y_b = \frac{1}{R}$$

$$Y_a = \frac{-j}{\omega L_2}$$

The total admittance is thus

$$\frac{1}{R} - \frac{j}{\omega L_2}$$

For a load angle $\theta$

$$\tan\theta = \frac{R}{\omega L_2}$$

or

$$L_2 = \frac{R}{\omega \tan\theta}$$

The slope of $Y_a$ is

$$\frac{d\{\mathrm{Im}(Y_a)\}}{d\omega} = \frac{1}{\omega^2 L_2}$$

Close to resonance $(\omega^2 C_1 L_1 - 1)^2$ tends to zero and the slope of $Y_b$ is

$$\frac{d\{\mathrm{Im}(Y_b)\}}{d\omega} = \frac{d}{d\omega}\left(\frac{1-\omega^2 C_1 L_1}{\omega C_1 R^2}\right) = -\frac{\omega^2 C_1 L_1 + 1}{\omega^2 C_1 R^2}$$

As $\omega^2 C_1 L_1 = 1$

$$\frac{d\{\mathrm{Im}(Y_b)\}}{d\omega} = \frac{-2}{\omega^2 C_1 R^2}$$

For slope cancellation

$$\frac{d\{\mathrm{Im}(Y_a)\}}{d\omega} = -\frac{d\{\mathrm{Im}(Y_b)\}}{d\omega}$$

and therefore

$$\frac{1}{\omega^2 L_2} = \frac{2}{\omega^2 C_1 R^2}$$

$$C_1 = \frac{2L_2}{R^2}$$

For resonance of $C_1$ and $L_1$

$$L_1 = \frac{1}{\omega^2 C_1}$$

The design criteria for the network are

$$L_2 = \frac{R}{\omega \tan\theta}, \quad C_1 = \frac{2L_2}{R^2}, \quad L_1 = \frac{1}{\omega^2 C_1}$$

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